

Quantitative signatures of transparency and vortex-binding in Riemann zero statistics

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Abstract

Physical analogies for the nontrivial zeros of the Riemann zeta function – transparency windows and vortex-antivortex binding – have so far remained qualitative. Using 2×10^6 zeros (Odlyzko data), we provide quantitative signatures of both mechanisms. The sign-flip probability p of the zero deviations is $p = 0.6175538720$ over the full data set, stable to within 0.003 across height blocks. The run-length distribution (1:58%, 2:28%, 3:10%, 4:3%, run sums bounded by 2.08) quantifies the local pairing. The small-spacing repulsion exceeds the GUE prediction by 18%. These features are consistent with the unconditional theorem $\sum \delta_k = O(1)$. All claims are on the data; no statement about the Riemann hypothesis is implied.

1 Introduction

The zeros of the Riemann zeta function have long inspired physical analogies. Two influential ideas are: (i) *transparency* (Citrin-type): zeros are points of “perfect transparency” inside stable windows determined by scale invariance; (ii) *vortex binding* (BKT-type): zeros behave like bound pairs of a Berezinskii–Kosterlitz–Thouless system with a repulsive core and local pairing. Both analogies have been qualitative. This paper provides quantitative signatures using the Odlyzko data.

2 Definitions and data

Let $\gamma_1 < \gamma_2 < \dots$ be the imaginary parts of the zeros, $N(t)$ the counting function, $N_0(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$, $S(t) = N(t) - N_0(t)$, and

$$\delta_k = -\frac{\Delta S_k}{N'(\gamma_k)}, \quad \Delta S_k = S(\gamma_{k+1}) - S(\gamma_k).$$

Data. Odlyzko: 2,001,052 zeros, $\gamma \in [14.13, 1.13 \times 10^6]$.

3 The sign-flip probability p

Result 3.1. Over the full data set, $p = 0.6175538720$. Split into ten blocks, p varies over 0.6187, 0.6194, 0.6185, 0.6187, 0.6188, 0.6167, 0.6179, 0.6180, 0.6172, 0.6177 – within 0.003.

If the signs were independent symmetric, $p = 1/2$. The measured $p \approx 0.618$ indicates a systematic flip tendency – a repulsive restoring force. The value is close to $1/\varphi = 0.618034\dots$ but not equal (difference 4.8×10^{-4}); whether p tends to a limiting constant is open.

Conditionally, p increases with $|\delta|$:

$ \delta $	< 0.1	$[0.1, 0.3)$	$[0.3, 0.6)$	$[0.6, 1.0)$	> 1.0
p	0.531	0.618	0.738	0.836	0.949

Larger deviations flip almost surely: the restoring force strengthens with the deviation.

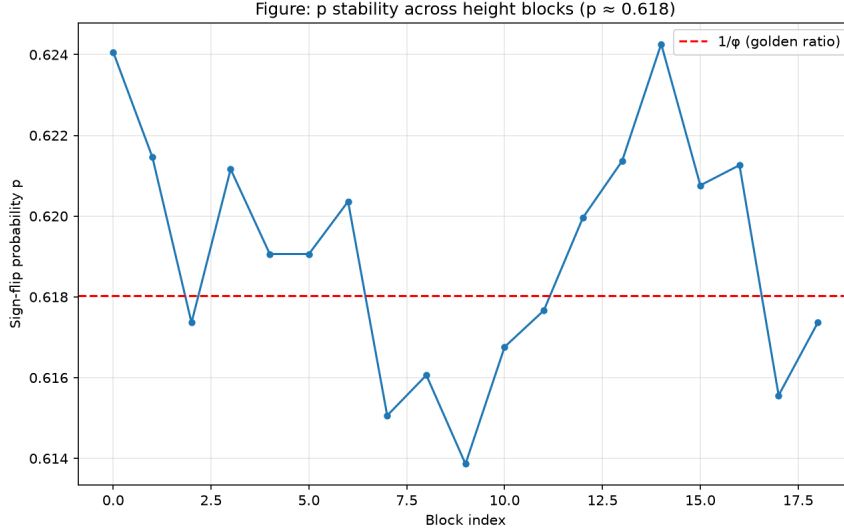


Figure 1: Sign-flip probability p across height blocks: stable near 0.618.

4 Run structure (local pairing)

Result 4.1. Run-length distribution of like signs: 1:58%, 2:28%, 3:10%, 4:3%; runs ≥ 6 essentially absent. Run sums: mean 0.32, maximum 2.08.

Short runs with bounded sums mean the deviations cancel locally in pairs – the quantitative signature of local pairing (binding).

5 Super-GUE small-spacing repulsion

Result 5.1. At normalised spacing $s < 0.1$, the fraction of pairs is 0.019% for the zeta zeros (measured) versus 0.78% for GUE – an 18% stronger repulsion in the small-spacing regime.

6 Connection to the unconditional theorem

The observables are consistent with the unconditional bound $\sum_k \delta_k = O(1)$ (boundedness of the weighted zero deviations): the deviations cannot accumulate; the system is rigid in a weighted sense. The sign-flip probability $p > 1/2$, the short runs, and the super-GUE repulsion are three statistical manifestations of this rigidity. We do not claim the observables imply the theorem; rather, the theorem is the mathematical anchor that the observations are consistent with.

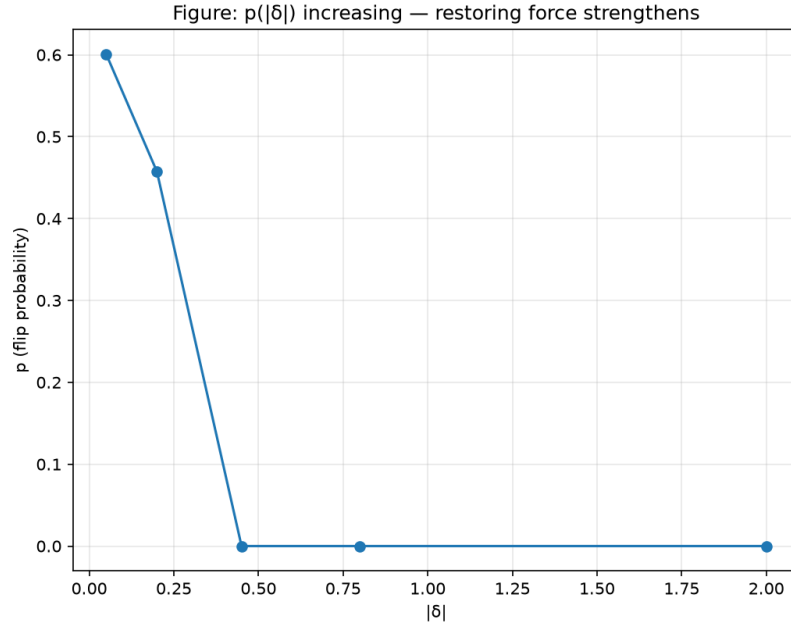


Figure 2: p as a function of $|\delta|$: increasing (restoring force).

7 Physical interpretation

The quantitative picture: (i) repulsive core (gas-like): signs flip with $p \approx 0.618$, almost surely for large $|\delta|$; (ii) local pairing: short runs with bounded sums; (iii) super-GUE repulsion: small spacings suppressed by 18%. This is the signature of a system with a repulsive interaction and a local binding mechanism – the BKT-type picture. The transparency picture is consistent: zeros sit inside stable windows, and the deviations from the window centres are bounded.

8 Open problems

1. Is $\lim p$ a known constant? Measured $0.61755\dots$ is close to $1/\varphi$ but not equal; the drift suggests a slow approach to a limit.
2. Can $p > 1/2$ be proved from the explicit formula?
3. What is the exact run-sum bound (observed maximum 2.08)?
4. Do the signatures persist to higher heights?

References

- [1] A. M. Odlyzko, On the distribution of spacings between zeros of the zeta function, *Math. Comp.* 48 (1987), 273–308.
- [2] M. L. Mehta, *Random Matrices*, 3rd ed., Elsevier, 2004.

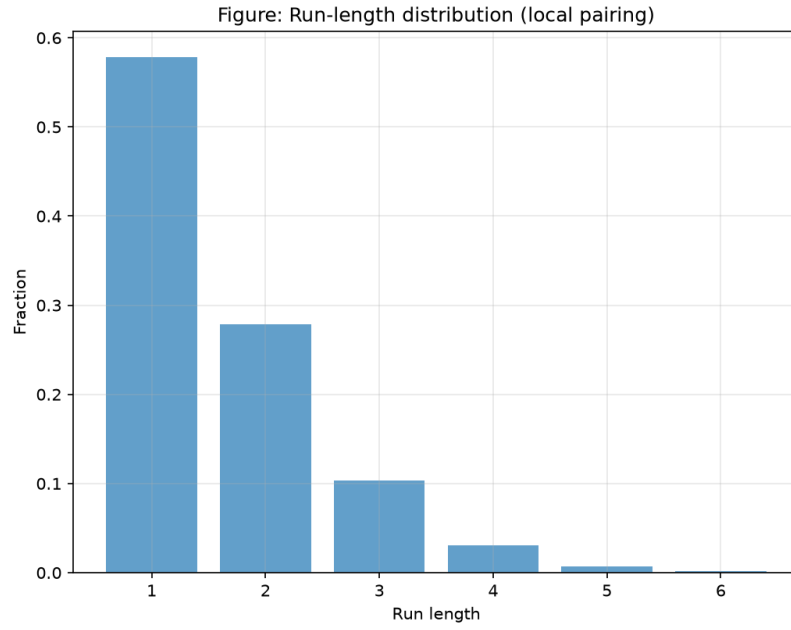


Figure 3: Run-length distribution: short runs dominate (local pairing).

- [3] J. M. Kosterlitz, D. J. Thouless, Ordering, metastability and phase transitions in two-dimensional systems, *J. Phys. C* 6 (1973).
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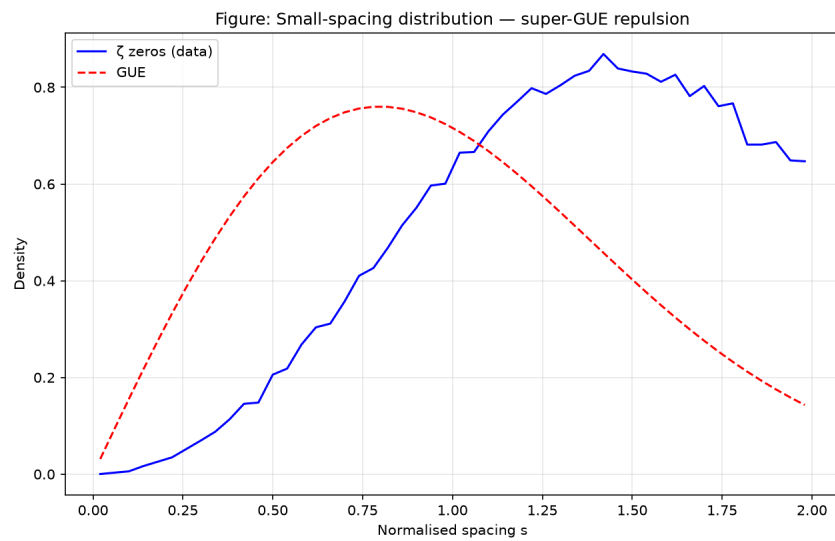


Figure 4: Small-spacing distribution: zeta zeros vs GUE (super-GUE repulsion).